

Generation Interconnection System Impact Study Report

For Queue Project Q12-0172

Morgantown Energy Facility 345 kV
December 28, 2014

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q12-0172. The Interconnection Customer proposes a 370.6 MW wind facility interconnecting at the Morgantown Energy Facility 345 kV substation in the West Virginia area. The request was received on November 08, 2012. The study evaluated system intact and single contingency conditions using the regional planning case, including 106 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 12 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$34,733,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

For the loss of SANTEE COOPER / GREENLAND GAP 345 KV, the ALLIGATOR 500-345 kV Auto loads to 138.7% of its 544 MVA rating. Project DFAX: 7.1%.

RIDGELEY / AMERICAN 345 KV Line: 125.4% loading (1,263 MVA) following the outage of the THIELE KAOLIN REEDY CRK / E DEVON 345 KV L; DFAX = 8.5%.

For the loss of AUBURN RD / ACEC DELLWOOD 345 KV line, the AUBURN ROAD 500/345 kV Transformer Bank loads to 121.0% of its 485 MVA rating. Project DFAX: 26.0%.

POTLATCH / CAVENDISH 138 KV: 117.2% loading (171 MVA) following the outage of the PRIDES CORNER / AUBUN RD 345 KV; DFAX = 6.7%.

INTERNATIONAL PAPER VALLIANT OK / URBAN 345 KV: 115.1% loading (1,276 MVA) following the outage of the URBAN / FRONT ROYAL 345 KV L; DFAX = 8.0%.

POTLATCH / GANS 138 KV: 113.9% loading (181 MVA) following the outage of the AUBUN ROAD / PRIDES CORNER 345 KV Line; DFAX = 6.7%.

SPINDLE / RIDGELEY 345 KV Line: 111.6% loading (1,495 MVA) following the outage of the E DEVON / THIELE KAOLIN REEDY CREEK 345 KV; DFAX = 8.9%.

For the loss of AUBURN RD / PRIDES CORNER 345 KV line, the PIT 7 TAP SUBSTATION / GANS 138 KV loads to 108.2% of its 196 MVA rating. Project DFAX: 6.7%.

The WESTWARD / CAVENDISH 138 KV is overloaded to 105.1% of its 187 MVA emergency rating for the loss of the AUBURN ROAD / PRIDES CORNER 345 KV Line. The project contributes 6.7% DFAX (24.8 MW) to this overload.

The YOUNGWOOD / WOODLAND TAP 345 KV is overloaded to 104.9% of its 1,221 MVA emergency rating for the loss of the SOUTHEAST VICKSBURG / ALLEGHENY ENERGY UNITS 3 4 & 5 500 KV L. The project contributes 19.1% DFAX (70.9 MW) to this overload.

For the loss of TEXAS EASTERN 5 / RONCO 500 KV, the YOUNGWOOD / UNIV OF NC CHAPEL HILL COGEN FACILITY 345 KV line loads to 103.3% of its 1,168 MVA rating. Project DFAX: 21.2%.

For the loss of INTERNATIONAL PAPER VALLIANT OK / URBAN 345 KV L, the URBAN / FRONT ROYAL 345 KV Line loads to 101.7% of its 1,270 MVA rating. Project DFAX: 7.0%.

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ALLIGATOR 500 kV/345 kV Auto	Replace transformer with higher-rated unit	\$26,748,000	13% (\$3,488,000)
AMERICAN / RIDGELEY 345 KV	Rebuild line with higher-capacity conductor	\$20,735,000	12% (\$2,490,000)
AUBURN ROAD 500/345 kV Transformer Bank	Replace transformer with higher-rated unit	\$20,211,000	29% (\$5,915,000)
POTLATCH / CAVENDISH 138 KV	Rebuild line with higher-capacity conductor	\$3,227,000	18% (\$587,000)
INTERNATIONAL PAPER VALLIANT OK / URBAN 345 KV Line	Rebuild line with higher-capacity conductor	\$69,812,000	13% (\$9,280,000)
POTLATCH / GANS 138 KV	Rebuild line with higher-capacity conductor	\$3,483,000	18% (\$633,000)
SPINDLE / RIDGELEY 345 KV Line	Rebuild line with higher-capacity conductor	\$29,827,000	12% (\$3,537,000)
PIT 7 TAP SUBSTATION / GANS 138 KV	Reconductor line	\$2,789,000	18% (\$507,000)
WESTWARD / CAVENDISH 138 KV	Reconductor line	\$2,033,000	18% (\$369,000)
YOUNGWOOD / WOODLAND TAP 345 KV	Reconductor line	\$25,957,000	12% (\$3,054,000)
YOUNGWOOD / UNIV OF NC CHAPEL HILL COGEN FACILITY 345 KV line	Reconductor line	\$7,248,000	19% (\$1,380,000)
URBAN / FRONT ROYAL 345 KV	Reconductor line	\$25,022,000	14% (\$3,493,000)
	Total allocated to project		\$34,733,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.